🛡️ Sentiment Guard v3.0

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Core Architecture

The Sentiment Guard Proof of Concept (PoC) AML model uses a Random Forest Classifier as its primary predictive engine. This ensemble learning method was chosen for its high accuracy in detecting non-linear fraud patterns and its resilience against "noisy" data common in financial transaction logs.

🔍 Intelligence Features

The "Brain" of the system evaluates entities across 43 distinct feature vectors, including:

- **Behavioral Velocity:** Tracking the speed and frequency of transactions to detect automated "structuring."
- **Relationship Forensic:** Analyzing the distance between account originators and destinations to identify "money mule" clusters.
- **Adverse Media Sentiment:** Real-time integration with global news feeds to cross-reference transactions with negative jurisdictional or entity-based press.

🛡️ Risk Scoring Logic

The model outputs a probability score (0-100%) which is visualized via the AI Risk Verdict gauge:

🟢 0% - 30% (Low Risk): Consistent with standard consumer or corporate behavior.

🟡 31% - 70% (Medium Risk): Exhibits behavioral anomalies; requires investigator review.

🔴 71% - 100% (High Risk): Matches known money laundering archetypes (Layering/Smurfing). Immediate escalation to SAR (Suspicious Activity Report) recommended.

🔍 Explainable AI (XAI) & Transparency

To move beyond "Black Box" predictions, Sentiment Guard v3.0 integrates a custom XAI Risk Driver module. This allows investigators to see the specific behavioral triggers behind every AI Risk Verdict.

- **Risk Attribution:** The dashboard breaks down the top 5 contributing factors (e.g., Transaction Velocity, Foreign Links, or Amt Scale) for the selected entity.
- **Investigative Efficiency:** By providing the "Why" behind the "What," the model helps analysts quickly verify AI findings and reduces the time required to draft SAR (Suspicious Activity Reports).
- **Regulatory Alignment:** This transparent approach aligns with emerging global standards for AI accountability in financial services.

📈 Adaptive Learning (PoC Capability)

Unlike static systems, this PoC features a Dynamic Ingestion Layer. By using the "Feed the Brain" portal, investigators can upload supplemental CSV data, allowing the system to immediately map new entities into the existing Relationship Network for real-time forensic analysis.

🛠️ Technical Specifications

Component	Specification	Description
Model Type	Random Forest	Ensemble classifier with 100+ decision trees.
Input Shape	1 x 43	Single transaction record with 43 pre-processed features.
Input Type	Float64	Numerical weights representing behavioral heuristics.

Component	Specification	Description
Processing Engine	Scikit-Learn	Industry-standard library for predictive data analysis.
Data Format	UTF-8 CSV	Standardized ingestion for base and supplemental data.
Serialization	Joblib	High-efficiency persistence for the trained model "brain."

📦 System Dependencies

To run Sentiment Guard v3.0, the following Python libraries must be installed:

bash

Core Framework streamlit # Interactive UI and Dashboard pandas # Data manipulation and cleaning

Visualization plotly # Gauge charts and choropleth maps networkx # Graph theory and network structure pyvis # Interactive HTML network rendering

AI & Data scikit-learn # Machine Learning model execution joblib # Model loading and serialization gnews # Real-time adverse media scanning

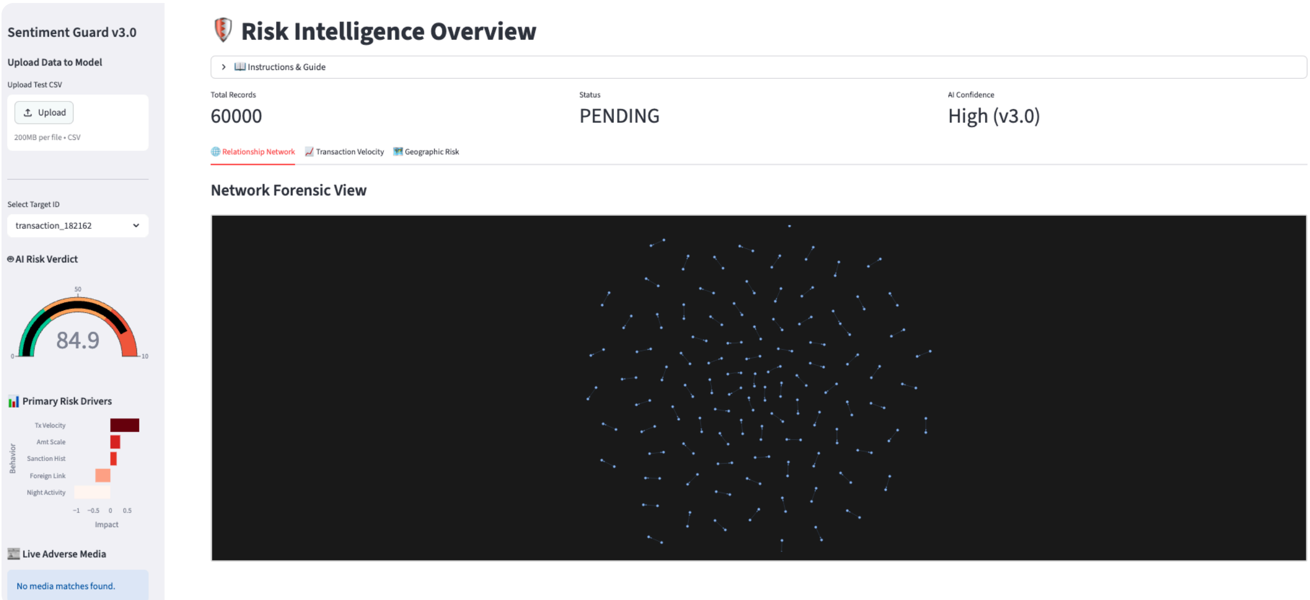
⚙️ Hardware & Deployment

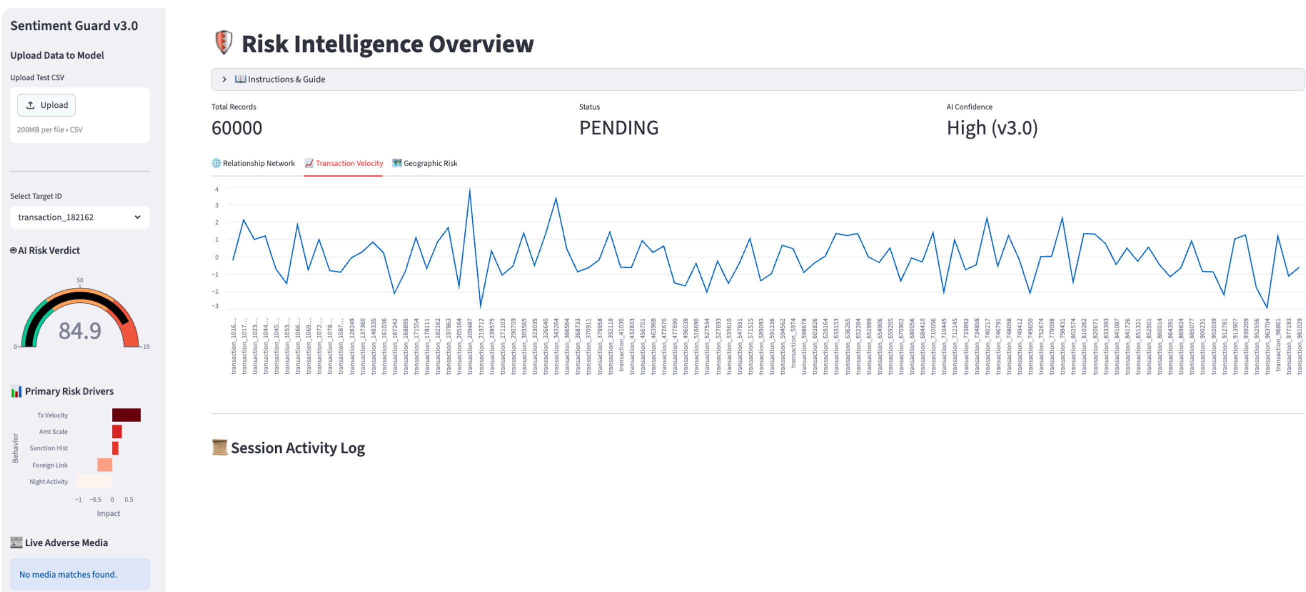
Hosting: Designed for Streamlit Community Cloud or local Docker containers.

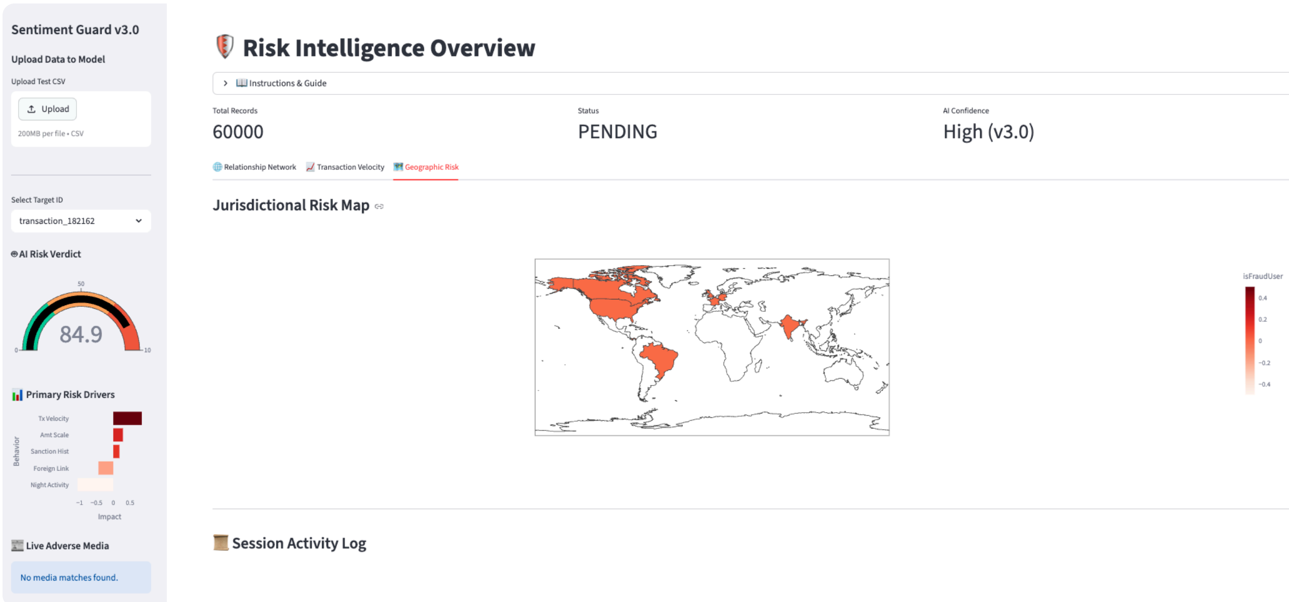
Latency: AI Risk Verdict generation in <200ms per entity.

Memory: High-efficiency caching allows for smooth handling of datasets up to 100k+ records on standard hardware.

📺 Screenshots






Sentiment Guard v3.0
Upload Data to Model
Upload Test CSV
Upload
200MB per file • CSV
Select Target ID
transaction_182162
AI Risk Verdict
84.9
Primary Risk Drivers
Tx Velocity
Amt Scale
Sanction Hist
Foreign Link
Night Activity
Behavior
Impact
Live Adverse Media
No media matches found.
Risk Intelligence Overview
Instructions & Guide
Total Records
60000
Status
PENDING
AI Confidence
High (v3.0)
Relationship Network
Transaction Velocity
Geographic Risk
Jurisdictional Risk Map
isFraudUser
Session Activity Log